



Modelling Financial Markets

Chapter 19: Review and Project Presentations



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Acronyms used in this material (1/2)

- ▣ **ADF**: Augmented Dickey–Fuller (test)
- ▣ **AI**: Artificial Intelligence
- ▣ **AR**: AutoRegressive (model); in event studies: Abnormal Return
- ▣ **ASE**: Academia de Studii Economice din București — *the Bucharest University of Economic Studies*
- ▣ **BET**: Bucharest Exchange Trading (index)
- ▣ **BET-TR**: BET Total Return (index)
- ▣ **BH**: Benjamini–Hochberg (procedure)
- ▣ **BNR**: Banca Națională a României — *the National Bank of Romania*
- ▣ **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- ▣ **CAPM**: Capital Asset Pricing Model
- ▣ **CI**: Confidence Interval
- ▣ **COVID-19**: COronaVirus Disease 2019
- ▣ **CV**: Cross-Validation
- ▣ **DCC**: Dynamic Conditional Correlation
- ▣ **DM**: Diebold–Mariano (test)
- ▣ **DQ**: Dynamic Quantile (test)
- ▣ **ES**: Expected Shortfall
- ▣ **ETF**: Exchange-Traded Fund
- ▣ **EVT**: Extreme Value Theory
- ▣ **FDR**: False Discovery Rate
- ▣ **FHS**: Filtered Historical Simulation
- ▣ **FRED**: Federal Reserve Economic Data
- ▣ **FTSE**: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ **FWER**: Family-Wise Error Rate
- ▣ **FX**: Foreign eXchange

Acronyms used in this material (2/2)

- ▣ **FZ0**: Fissler–Ziegel loss, zero-homogeneous version
- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GARCH-EVT**: GARCH filter combined with Extreme Value Theory for the tail
- ▣ **GBM**: Geometric Brownian Motion
- ▣ **GJR**: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- ▣ **GRS**: Gibbons–Ross–Shanken (test)
- ▣ **GSADF**: Generalised Sup Augmented Dickey–Fuller (test)
- ▣ **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ **HAR**: Heterogeneous AutoRegressive (model)
- ▣ **HS**: Historical Simulation
- ▣ **IGARCH**: Integrated GARCH
- ▣ **IQR**: InterQuartile Range
- ▣ **LRMES**: Long-Run Marginal Expected Shortfall
- ▣ **MCS**: Model Confidence Set
- ▣ **MDE**: Minimum Detectable Effect
- ▣ **ML**: Machine Learning
- ▣ **MSE**: Mean Squared Error
- ▣ **QLIKE**: Quasi-LIKElihood (loss function)
- ▣ **QMLE**: Quasi-Maximum Likelihood Estimator
- ▣ **SADF**: Sup Augmented Dickey–Fuller (test)
- ▣ **SE**: Standard Error
- ▣ **SML**: Security Market Line
- ▣ **SRISK**: Systemic RISK (capital shortfall measure)
- ▣ **US**: United States
- ▣ **VIX**: CBOE Volatility Index
- ▣ **VOLARE**: VOLatility Archive for Realized Estimates (database)

Chapter Route and Outcomes

Route of this chapter

- Part I: the course in one picture, a review of [Chapters 0–18](#), six pitfalls, and one integrated case study, from raw BET prices to a backtested value-at-risk (VaR) at 1%
- Part II: the team project, AI in the project, a research case study on nonstandard errors, one pass of the AI discovery loop
- Technical detail sits in the [Appendix](#) at the end

After this chapter you can

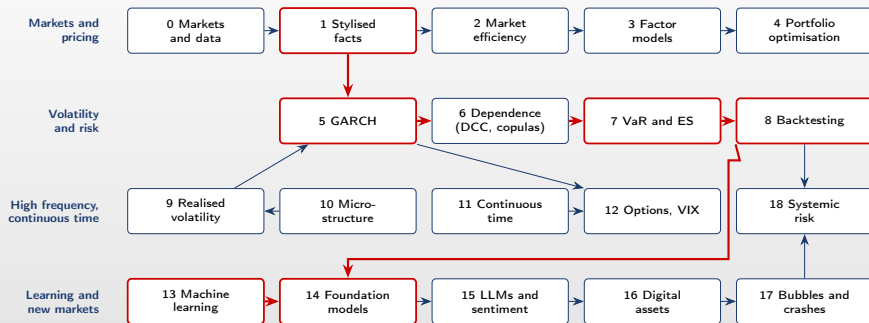
- run one question from data to a backtested risk forecast with valid inference
- recognise the six pitfalls in your own code
- write a project plan with replication, identification, power and a specification grid



Bucharest University of Economic Studies (ASE),
Piața Romană

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The Course in One Picture



- Red: the main thread, from stylised facts ([Chapter 1](#): heavy tails, volatility clustering, almost no linear predictability) to risk forecasts we can test ([Chapters 5–8](#): GARCH gives σ_{t+1} , VaR and ES the loss, backtesting says whether to trust it)
- [Chapters 9–12](#) measure volatility better (intraday data, option prices) and in continuous time
 - ▶ [Chapters 13–15](#) bring flexible models, always judged with the tools of [Chapter 8](#)
- [Chapters 16–18](#) apply the same toolkit to crypto, bubbles and the whole financial system
 - ▶ Special [Chapter 20](#): realised volatility with path signatures on the VOLARE data

Five Questions for Every Model

Before estimating

- **Data:** which series, which calendar, which adjustments (dividends, splits, holidays)?
- **Model**
 - ▶ which stylised fact does it capture?
 - ▶ which does it ignore?
- **Estimation:** on which window, with which information at time t ?

After estimating

- **Evaluation:** out of sample, with a loss that matches the target (QLIKE, FZ0) and the right diagnostic tests (e.g. Kupiec)?
- **Uncertainty:** robust standard errors, bootstrap, a test against a simple benchmark?
- If one answer is missing, the result is not yet a finding

Chapters 0–2: Markets, Returns and Efficiency

□ Chapter 0, Financial Markets in 2026

- ▶ Markets combine exchanges, over-the-counter dealers, exchange-traded funds (ETFs), algorithms and blockchains
- ▶ 2015–2026: SPY (S&P 500 ETF, dividends reinvested) 13.7% a year with 17.6% volatility; Bitcoin 60.6% a year with 66.7%
- ▶ BET, July 2007 to February 2009: a drawdown of -82.5%

□ Chapter 1, Stylised Facts and Returns

- ▶ Returns are uncorrelated but not independent: their magnitude is predictable, but their sign is largely unpredictable
- ▶ BET, Hill index of losses (top $k = 153$): $\hat{\alpha} = 2.39$, block-bootstrap 95% CI [2.12, 3.03] (α : tail index, the probability of a loss above x decays like $x^{-\alpha}$): the fourth moment is likely infinite
- ▶ Days beyond 5σ are more than 1000 times more frequent than the Normal distribution predicts

□ Chapter 2, Market Efficiency

- ▶ Efficiency: prices fully reflect the information set
- ▶ A test of abnormal-return predictability is a joint test of efficiency and the expected-return model [Fama \(1970\)](#)
- ▶ Share of rolling windows that reject the random walk: S&P 500 6.4%, BET 22.4%, Bitcoin 0.0%
- ▶ Statistical rejection is not exploitable profit

Chapters 3–5: Asset Pricing, Portfolios and GARCH

□ Chapter 3, Factor Models and Asset Pricing

- ▶ Expected returns should reward only exposure to systematic factors
- ▶ 25 size and book-to-market portfolios: SML (security market line) slope -4.4% , GRS (joint test that all intercepts are zero) = 4.18: the CAPM is rejected
- ▶ Sector ETFs: the first principal component (the market) explains 66% of the variance

□ Chapter 4, Portfolio Optimisation

- ▶ Markowitz asks the right question, but estimation error dominates the answer [Markowitz \(1952\)](#)
- ▶ 60-month samples: median in-sample Sharpe 1.57, median true Sharpe 0.30
- ▶ 96% of the estimated tangency portfolios are truly worse than $1/N$

□ Chapter 5, Conditional Volatility: GARCH

- ▶ Volatility clusters, so today's shock predicts tomorrow's variance [Engle \(1982\)](#), [Bollerslev \(1986\)](#)
- ▶ S&P 500, 2000–2026: persistence $\alpha + \beta = 0.9947$ (α, β : GARCH(1,1) weights of the last shock and of the last variance), a half-life $\ln 0.5 / \ln(\alpha + \beta)$ of 131 days
- ▶ BET: profile-likelihood 95% CI for $\alpha + \beta$ is $[0.975, 1]$ (IGARCH not rejected): the half-life is at least 27 days, with no finite upper bound

Chapters 6–8: Dependence, Risk Measures and Backtesting

□ Chapter 6, Multivariate Volatility and Dependence

- ▶ Correlations move with the macro regime; diversification is not guaranteed
- ▶ SPY–TLT (long US Treasury bond ETF): ρ from -0.40 to 0.11 after 2022, so a 60/40 portfolio went from 10.4% to 13.2% volatility
- ▶ Bitcoin–SPY DCC correlation: 0.04 before 2020, 0.31 since

□ Chapter 7, Value-at-Risk and Expected Shortfall

- ▶ VaR is a quantile of the loss, ES the average loss beyond it; ES is coherent
- ▶ S&P 500, daily: historical $\text{VaR}_{1\%} = 3.45\%$, $\text{ES}_{1\%} = 5.01\%$
- ▶ The Normal distribution gives only 2.80%: heavy tails matter

□ Chapter 8, Backtesting and Evaluating Risk Forecasts

- ▶ Test the number and the timing of breaches: coverage and independence [Kupiec \(1995\)](#), [Christoffersen \(1998\)](#)
- ▶ S&P 500, VaR 1%, 2007–2026: Normal 2.70%, HS 1.67%, FHS 1.23%, GARCH-EVT 1.13% breaches
- ▶ A GARCH filter plus a flexible tail wins in all four markets

Chapters 9–11: High Frequency, Liquidity and Continuous Time

Chapter 9, Realised Volatility

- ▶ Intraday prices measure daily variance (RV , realised variance)
- ▶ The log-HAR (heterogeneous autoregressive) model captures its long memory Corsi (2009)
- ▶ SPY: kurtosis 15.73 for open-to-close returns, 2.65 after dividing by $\sqrt{RV_t}$
- ▶ log-HAR beats GARCH(1,1)- t in QLIKE by 10% for SPY ($p = 0.021$) and 22% for Bitcoin

Chapter 10, Market Microstructure

- ▶ Prices form in limit order books; the spread pays for processing, inventory and adverse selection
- ▶ Amihud ratio (basis points, bp, per USD 1 million): SPY 0.0016, Banca Transilvania 24.7, Transelectrica 1024.0
- ▶ SPY: the opening 5-minute bar moves 2.06 times as much as a midday bar

Chapter 11, Continuous-Time Models

- ▶ Brownian motion is the limit of random walks; Itô's lemma solves the price equations
- ▶ S&P 500 against geometric Brownian motion: 62 days beyond ± 4 standard deviations, against 0.59 expected
- ▶ Hill index of S&P 500 losses: 2.87 [2.07, 3.67] (Chapter 16): the fourth moment is likely infinite, unlike the Normal log returns of GBM

Chapters 12–14: Options, Machine Learning and Foundation Models

□ Chapter 12, Options and the Volatility Surface

- ▶ Black–Scholes prices options by hedging; implied volatility is the market's view of σ Black & Scholes (1973)
- ▶ 1990–2026: mean VIX 19.4, mean realised volatility over the next month 15.3
- ▶ The variance risk premium is positive on 86% of days, but not in crashes

□ Chapter 13, Machine Learning in Finance

- ▶ Flexible models, but little signal and much noise; the finance-specific part is the process López de Prado (2018)
- ▶ Shuffled K-fold reports 69.4% accuracy on pure noise: leakage
- ▶ S&P 500: the fractionally differenced series is stationary at $d^* = 0.25$; no model beats “always up”

□ Chapter 14, Deep Learning and Time-Series Foundation Models

- ▶ LSTMs (long short-term memory networks) and Transformers learn from sequences; foundation models forecast zero-shot
- ▶ SPY realised volatility, QLIKE: Chronos-2 0.231, log-HAR 0.234, GARCH-t 0.283; the first two are not significantly different
- ▶ Chronos-2 VaR 1% without recalibration on the S&P 500: 1.67% breaches

Chapters 15–16: Text and Digital Assets

□ Chapter 15, LLMs and Sentiment Analysis

- ▶ From dictionaries to FinBERT and large language models (LLMs): text becomes a number
- ▶ Qwen 7B vs 1.5B on financial headlines: +21.5 percentage points in accuracy, McNemar $p < 0.001$
- ▶ A headline-sentiment strategy: Sharpe +0.63 before and −1.48 after costs; break-even cost 1.5 bp per trade

□ Chapter 16, Digital Assets and DeFi

- ▶ Cryptocurrencies form a distinct asset class in the feature space of [Pele et al. \(2023\)](#), where a tail factor separates them
- ▶ Their Hill tail indices are not significantly lower than those of equities or gold ([Chapter 16](#))
- ▶ Bitcoin–S&P 500 link: sup-Wald 20.2 (the largest Wald break statistic over candidate dates; 5% critical value 11.6), break estimated on 9 Mar 2020, not at the 2024 ETF launch
- ▶ Hill index of losses: Bitcoin 2.73 [2.12, 3.33], S&P 500 2.87 [2.07, 3.67]: statistically indistinguishable

Chapters 17–18: Bubbles and Systemic Risk

□ Chapter 17, Bubbles and Crashes

- ▶ A bubble is a price path that grows faster than fundamentals can justify
- ▶ Bitcoin 2014–2026 (weekly): GSADF (the largest right-tailed unit-root statistic over all windows) 3.85 against a 95% critical value of 2.27: explosive episodes are detected
- ▶ One vs two Markov regimes for Bitcoin: likelihood-ratio statistic $LR = 123.2$ against a parametric-bootstrap 95% critical value of 11.8 ($p = 0.005$)

□ Chapter 18, Systemic Risk and Networks

- ▶ Systemic risk: losses of one institution that spread to the whole system
- ▶ ΔCoVaR 1% (the change in the system's VaR 1% when a bank moves from its median to distress) of eleven banks: 4 of 25 within-region pairs differ at 5%, 0 after Holm
- ▶ Overlap of intervals alone would have ranked none
- ▶ Simulated LRMEs (22 days, market below -10%): JPMorgan 10.6% today, 23.7% with the information of 31 March 2020

Pitfalls of Design: Look-Ahead Bias and Overfitting

- ▣ **Look-ahead bias:** any parameter, threshold or scaling estimated on the full sample leaks the future
 - ▶ Examples: full-sample GARCH parameters, full-sample standardisation, labels that overlap the test set; the case study measures the effect on BET
 - ▶ Chapter 13: shuffled K-fold reports 69.4% accuracy on pure noise
 - ▶ Remedy: rolling or expanding windows; purging and embargo; write the information set $\mathcal{F}_{t-1}$ next to every forecast
- ▣ **Overfitting:** the in-sample fit rewards noise that will not repeat
 - ▶ Chapter 4: in-sample Sharpe 1.57, true Sharpe 0.30 for the same portfolios
 - ▶ Backtest overfitting: trying many strategies and keeping the best Bailey et al. (2017)
 - ▶ The deflated Sharpe ratio corrects for the number of trials and for non-normal returns Bailey & López de Prado (2014)
 - ▶ Remedy: fix the design before looking at the test period
 - ▶ Report every model you tried
 - ▶ Compare with a simple benchmark (1/N, historical simulation, random walk, log-HAR)

Pitfall of Design: Multiple Testing

- Chapter 3: with 300 useless factors, on average 15.0 pass $|t| > 1.96$, and the best reaches $|t| \approx 3.06$
 - ▶ This is why the factor literature asks for $|t| > 3$ Harvey, Liu & Zhu (2016)
- Calendar anomalies, trading rules, model specifications: each extra test adds a chance of a false discovery
- Remedy: count your tests and correct
 - ▶ Bonferroni and Holm control the chance of any false discovery (FWER); Benjamini–Hochberg controls their expected share (FDR)
 - ▶ For forecasts: the model confidence set (MCS) instead of many pairwise tests Hansen, Lunde & Nason (2011)

Pitfall of Data: Aligning Returns Instead of Prices

- ❑ Bitcoin trades every day, SPY only on weekdays
 - ▶ Wrong: compute each return on its own calendar, then keep the common days: Bitcoin's weekend moves disappear
 - ▶ Right: keep the common days of the **prices**, then compute returns: Monday covers Friday to Monday for both
- ❑ Bitcoin, 2020–2026, on the common days: cumulative return 1 025% (right) against 680% (wrong)
 - ▶ Monday standard deviation 5.04% (right) against 3.83% (wrong): the wrong alignment hides weekend risk
- ❑ Remedy: align prices on common days first; one series alone stays on its own calendar

Pitfall of Inference: Heavy Tails Ignored

- ▣ Classical standard errors assume independence and finite fourth moments
 - ▶ BET losses: Hill tail index $\hat{\alpha} = 2.39$, block-bootstrap 95% CI [2.12, 3.03] [Hill \(1975\)](#)
 - ▶ α : the probability of a loss above x decays like $x^{-\alpha}$, so moments of order $\geq \alpha$ are infinite: with $\alpha < 4$ the population kurtosis is infinite
 - ▶ So the sample kurtosis (11.1) estimates nothing, and a bootstrap interval for it is not valid [Athreya \(1987\)](#): report the tail index instead
- ▣ BET GARCH persistence $p = \alpha + \beta$ (here α, β : the GARCH(1,1) parameters): the Wald CI [0.974, 1.007] crosses the stationarity boundary $p = 1$, where the Normal approximation fails [Andrews \(1999\)](#)
 - ▶ Profile-likelihood CI (all p whose likelihood-ratio statistic, maximising over the other parameters, is below the critical value) inside the parameter space $p \leq 1$: [0.975, 1] (IGARCH, $p = 1$, not rejected against the boundary critical value 2.71, the 5% point of a 50:50 mixture of 0 and χ_1^2), so the half-life is at least 27 days and has no finite upper bound
- ▣ Remedy: robust (HAC) errors, block bootstrap where it is valid, tail indices instead of kurtosis, boundary-aware intervals

Pitfall of Evaluation: a Loss That Cannot Rank the Target

- Variance forecasts are compared with a noisy proxy (r_t^2 or RV_t)
 - ▶ With a conditionally unbiased proxy, MSE and QLIKE rank forecasts as the true variance would
 - ▶ QLIKE depends only on the ratio proxy/forecast, so volatile days weigh less than under MSE [Patton \(2011\)](#)
 - ▶ [Chapter 9](#): log-HAR beats GARCH in QLIKE; the MSE comparison is inconclusive
- ES alone has no strictly consistent scoring function on the usual broad classes of distributions
 - ▶ The pair (VaR, ES) has one, under mild conditions [Fissler & Ziegel \(2016\)](#)
 - ▶ Use FZ0 with Diebold–Mariano tests [Diebold & Mariano \(1995\)](#); expect finite-sample limits on precision [Pele & Mazurencu-Marinescu-Pele \(2026\)](#)
- Remedy: choose the loss before the comparison, and state it in the project report

One Question, Four Chapters

The question

- How much can a BET index position lose tomorrow, with probability 1%?
- Which forecast of that loss would we have trusted in real time, 2005–2026?

Data

- BET index, official daily closing values, 6 Jan 2000 – 18 Sep 2026: 6 670 daily log returns
- Duplicate records (rows identical to the previous one) and weekend-dated rows removed
 - ▶ Trading days with an unchanged close kept

The five steps

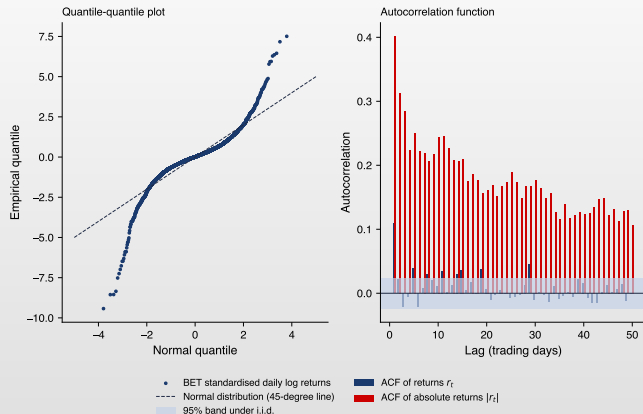
- **Chapter 1:** stylised facts, with robust inference
- **Chapter 5:** GARCH(1,1) with Student-*t* innovations
- **Chapter 7:** one-day VaR 1% from four models
- **Chapter 8:** coverage, independence and Basel zones
- Formal comparison of the four models: FZ0 loss, DM tests, model confidence set



The Stock Exchange Palace, Bucharest (2023)

Photo: Chainwit. (2023); CC BY 4.0; Wikimedia Commons

Step 1: Stylised Facts of the BET



- Heavy tails (left, quantiles against the Normal distribution)
 - ▶ Hill tail index of losses $\hat{\alpha} = 2.39$, 95% CI [2.12, 3.03] (block bootstrap, $k = 153$ largest losses)
 - ▶ 51 days beyond ± 4 s.d. vs 0.42 under the Normal; worst day -13.1% on 7 Jan 2009
- Volatility clustering (right)
 - ▶ Autocorrelation of $|r_t|$ at lag 1: 0.40, slow decay; Ljung-Box $Q(10)$ on r_t^2 : 2440 (5% critical value 18.3)
- A Normal, constant-volatility VaR: too low in stress, too high in calm periods

Step 1: Reading the Stylised Facts

- Hill tail index $\hat{\alpha}$: the tail of the losses decays like $x^{-\alpha}$
 - ▶ Moments of order $\geq \alpha$ are infinite: with $\hat{\alpha} < 4$ the kurtosis is not reported as an estimate
- Ljung–Box $Q(10)$: a joint test that the first 10 autocorrelations are zero
 - ▶ On r_t^2 it tests for volatility clustering; under H_0 , $Q(10) \sim \chi_{10}^2$
- Autocorrelation of the returns themselves: 0.11
 - ▶ Typical of a less liquid market ([Chapters 2, 10](#))

Step 2: the GARCH(1,1)- t Model

- Today's variance is a weighted sum of a constant, yesterday's squared shock and yesterday's variance [Bollerslev \(1986\)](#)

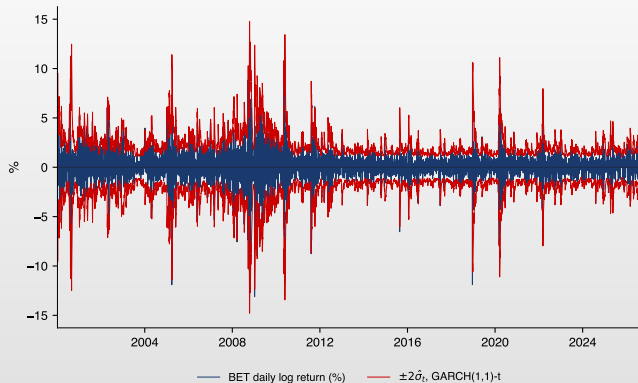
$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad \sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

- r_t : daily log return; μ : mean; ε_t : shock; σ_t : conditional volatility, known at $t - 1$; z_t : i.i.d. Student- t with ν degrees of freedom, scaled to variance 1
 - $\omega > 0, \alpha, \beta \geq 0$: α measures the reaction to news, β the memory; ν small: heavy tails ($\nu \rightarrow \infty$ gives the Normal distribution)
- Persistence $\alpha + \beta < 1$: covariance stationary; a variance shock decays by the factor $\alpha + \beta$ each day

$$h_{1/2} = \frac{\ln 0.5}{\ln(\alpha + \beta)}$$

- $h_{1/2}$: half-life, the number of days after which half of a variance shock is gone; $\alpha + \beta = 1$ (IGARCH): shocks never fade
- Estimated by maximum likelihood, robust (sandwich) standard errors; the next slide gives the BET estimates

Step 2: GARCH(1,1)- t Volatility



- BET daily log returns with $\pm 2\hat{\sigma}_t$ from a GARCH(1,1) with Student- t innovations, 2000–2026
 - ▶ Robust standard errors: $\hat{\alpha} = 0.192$ (0.025), $\hat{\beta} = 0.799$ (0.024), $\hat{\nu} = 5.01$: heavy tails even after filtering
- Persistence $\hat{\alpha} + \hat{\beta} = 0.9907$, a half-life of about 74 trading days
 - ▶ Profile-likelihood 95% CI [0.975, 1]: persistence cannot be bounded away from 1 (pitfall of inference)
- Peak daily volatility 7.39% on 15 Oct 2008; at the end of the sample 1.67%

Step 3: Four VaR 1% Forecasts (1/2)

- Unconditional, on the last 500 days: one quantile for the whole window
 - ▶ **HS** (historical simulation): the loss $L = -r$ exceeded on 1% of the last 500 days (empirical quantile)
 - ▶ **Normal**: $\text{VaR} = -(\hat{\mu} + \hat{\sigma} z_{0.01})$; $\hat{\mu}$, $\hat{\sigma}$: sample mean and standard deviation; $z_{0.01} = -2.326$: the 1% quantile of the standard Normal distribution
- Conditional, GARCH- t on the last 1000 days: the quantile scales with today's volatility forecast $\hat{\sigma}_t$

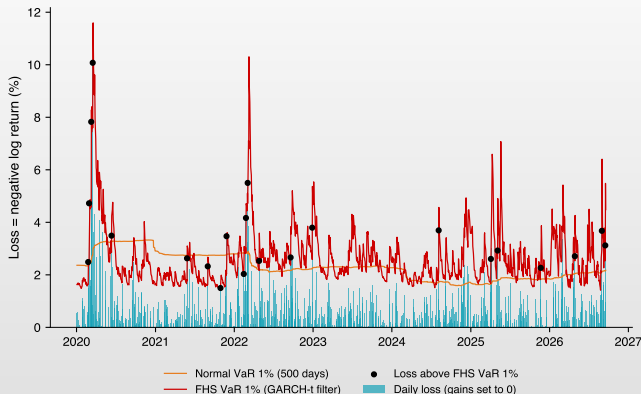
$$\text{GARCH-}t: \text{VaR}_t = -\left(\hat{\mu} + \hat{\sigma}_t t_{\nu}^{-1}(0.01) \sqrt{(\nu - 2)/\nu}\right), \quad \text{FHS: } \text{VaR}_t = -(\hat{\mu} + \hat{\sigma}_t \hat{q}_{0.01}(z))$$

- ▶ $t_{\nu}^{-1}(0.01)$: the 1% quantile of a Student- t with ν degrees of freedom; $\sqrt{(\nu - 2)/\nu}$ rescales it to variance 1
- ▶ **FHS** (filtered historical simulation): $\hat{q}_{0.01}(z)$, the empirical 1% quantile of the standardised residuals $z_t = (r_t - \hat{\mu})/\hat{\sigma}_t$, replaces the Student- t quantile [McNeil & Frey \(2000\)](#)

Step 3: Four VaR 1% Forecasts (2/2)

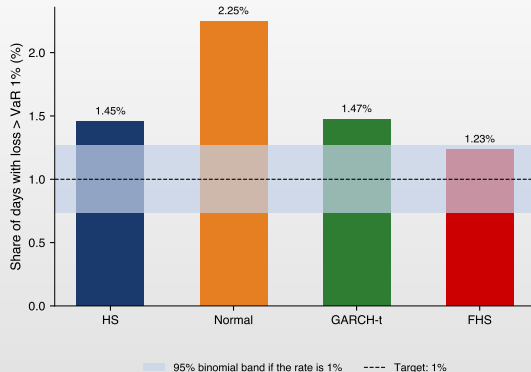
- Real-time design
 - ▶ Forecast for day t uses data up to $t - 1$ only
 - ▶ GARCH re-estimated every 21 trading days; $\hat{\sigma}_t$ updated daily
- Evaluation: 3 Jan 2005 – 18 Sep 2026, $T = 5\,432$ days
 - ▶ A breach (hit) on day t : the loss $-r_t$ exceeds VaR_t ; expected breaches at 1%: $0.01 T \approx 54$
- Unconditional models react only as the window rolls; conditional ones react through $\hat{\sigma}_t$, within days

Step 3: Losses and VaR 1%, 2020–2026



- Daily BET losses (gains set to 0), Normal and FHS VaR 1%; black dots: losses above the FHS VaR
- FHS reacts within days: it peaked at 11.6% on 18 Mar 2020, during the COVID-19 crash
 - ▶ The Normal VaR on a 500-day window moves only slowly
- 1 675 days, about 17 breaches expected: FHS 22, Normal 41
 - ▶ Normal breaches cluster in the first weeks of every stress episode
- Last forecast: FHS 4.48%, Normal 2.18%: the models disagree most when volatility changes

Step 4: Breach Rates



- Share of days with a loss above the VaR 1% forecast, 3 Jan 2005 – 18 Sep 2026, $T = 5\,432$
- Light-blue band: the 95% binomial range [0.74%, 1.27%] if the true rate is 1%
- Only FHS (1.23%) lies inside the band; the Normal model (2.25%) breaches more than twice as often as it should
- Next
 - ▶ are the differences significant?
 - ▶ are the breaches independent?

Step 4: the Three Backtests

- Kupiec coverage test [Kupiec \(1995\)](#): is the breach rate equal to $\alpha = 1\%$?

$$LR_{uc} = -2 \ln \frac{(1 - \alpha)^{T-x} \alpha^x}{(1 - \hat{\pi})^{T-x} \hat{\pi}^x} \sim \chi_1^2, \quad \hat{\pi} = x/T$$

- ▶ x : number of breaches in T days; $\hat{\pi}$: observed breach rate; $LR_{uc} = 0$ when $\hat{\pi} = \alpha$; reject at 5% when $LR_{uc} > 3.84$

- Christoffersen independence test [Christoffersen \(1998\)](#): is a breach today more likely after a breach yesterday?

$$LR_{ind} = -2 \ln \frac{(1 - \hat{\pi})^{n_{00}+n_{10}} \hat{\pi}^{n_{01}+n_{11}}}{(1 - \hat{\pi}_{01})^{n_{00}} \hat{\pi}_{01}^{n_{01}} (1 - \hat{\pi}_{11})^{n_{10}} \hat{\pi}_{11}^{n_{11}}} \sim \chi_1^2$$

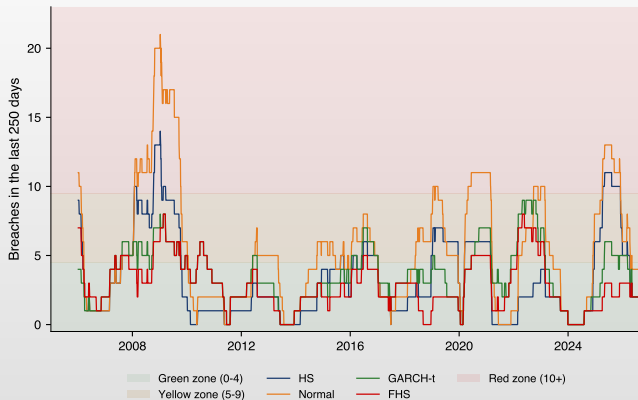
- ▶ n_{ij} : number of days in state j after a day in state i ($1 = \text{breach}$, $0 = \text{no breach}$)
- ▶ $\hat{\pi}_{01} = n_{01}/(n_{00} + n_{01})$, $\hat{\pi}_{11} = n_{11}/(n_{10} + n_{11})$: breach rates after a quiet day and after a breach
- ▶ Joint test: $LR_{cc} = LR_{uc} + LR_{ind} \sim \chi_2^2$ (5% critical value 5.99); the p -values p_{uc} , p_{ind} , p_{cc} come from these χ^2 distributions

Step 4: Coverage and Independence Tests

Model	Breaches	Rate	p_{uc}	p_{ind}	p_{cc}	Red zone
HS	79	1.45%	0.002	<0.001	<0.001	5.5%
Normal	122	2.25%	<0.001	<0.001	<0.001	18.6%
GARCH-t	80	1.47%	0.001	0.864	0.005	0.0%
FHS	67	1.23%	0.095	0.267	0.134	0.0%

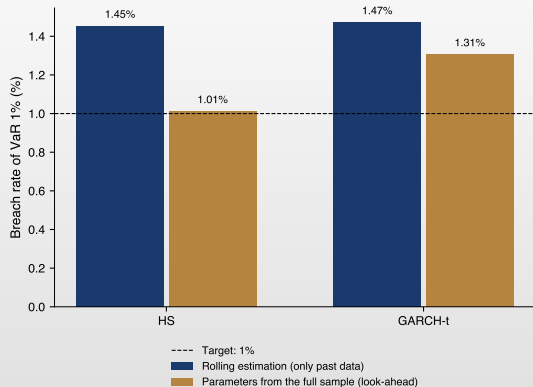
- p_{uc} : Kupiec coverage test; p_{ind} : Christoffersen independence test; p_{cc} : both jointly [Kupiec \(1995\)](#), [Christoffersen \(1998\)](#)
- Red zone: share of 250-day windows with 10 or more breaches (Basel traffic light)
- Only FHS is rejected by none of the three tests at 5%; HS and Normal also fail independence: their breaches cluster
- GARCH- t : independence is not rejected, but there are too many breaches: the Student- t tail is not heavy enough at 1%

Step 4: Breaches in the Last 250 Days



- Number of breaches in a moving 250-day window, with the Basel green, yellow and red zones
- The Normal model spends 18.6% of the windows in the red zone, HS 5.5%, the two GARCH-filtered models never

The Look-Ahead Trap in the Case Study



- Full-sample HS quantile: 1.01% breaches, by construction
- GARCH- t with full-sample parameters: 1.31% of days, with rolling estimation 1.47%
- Both look better than their real-time versions: the gap is the value of information nobody had
- Only the rolling result is evidence about a forecasting model

Step 5: Three Tools for a Formal Comparison (1/2)

- **Dynamic Quantile (DQ) test** [Engle & Manganelli \(2004\)](#): are the breaches predictable?
 - ▶ Regress $I_t - 0.01$ on a constant, the four lagged hits $I_{t-1}, \dots, I_{t-4}$ and VaR_t ; $I_t = 1$ if the loss exceeds VaR_t , else 0
 - ▶ DQ: Wald statistic that all six coefficients are zero, $DQ \sim \chi^2_6$ under a correct model
 - ▶ It detects breaches predictable from the past or from the level of VaR, which the Christoffersen test (one lag) can miss
- **FZ0 score** [Patton, Ziegel & Chen \(2019\)](#): one number per day that scores a (VaR, ES) forecast pair; lower is better

$$S(v, e; y) = -\frac{1\{y \leq v\}(v - y)}{\alpha e} + \frac{v}{e} + \ln(-e) - 1$$

- ▶ $\alpha = 2.5\%$; $v = -\text{VaR}$, $e = -\text{ES}$: the forecasts written as return quantiles ($e \leq v < 0$); y : the realised return; $1\{\cdot\}$: 1 if the condition holds, else 0
- ▶ Strictly consistent for (VaR, ES) [Fissler & Ziegel \(2016\)](#): on average the true pair has the lowest score

Step 5: Three Tools for a Formal Comparison (2/2)

▣ Comparing the FZ0 scores of two models

- ▶ Pairwise Diebold–Mariano tests [Diebold & Mariano \(1995\)](#): $t = \bar{d} / \sqrt{\hat{\Omega} / T}$, d_t the FZ0 difference of the two models on day t , $\hat{\Omega}$ its HAC long-run variance
- ▶ $t \approx N(0, 1)$ under equal accuracy
- ▶ Model confidence set at 10% [Hansen, Lunde & Nason \(2011\)](#): the set of models that contains the best one with probability 90%
- ▶ p_{MCS} : a model stays in the set while $p_{MCS} \geq 0.10$

▣ ES backtest

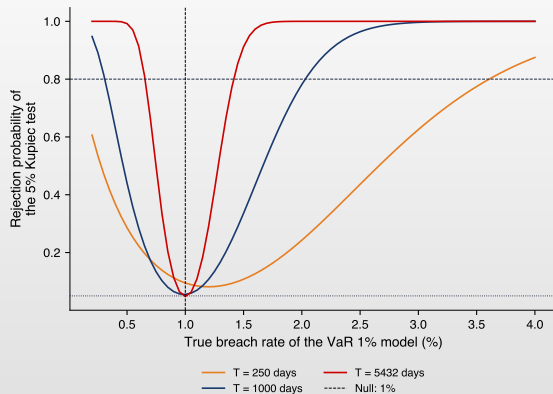
- ▶ Comparative, against HS as the standard model; red if the model is significantly worse, green if significantly better, yellow otherwise [Nolde & Ziegel \(2017\)](#)
- ▶ All forecasts use only data up to $t - 1$; 5 432 days, 3 Jan 2005 – 18 Sep 2026

Step 5: Results

Model	DQ	p_{DQ}	FZ0	p_{MCS}	ES zone
HS	192.1	<0.001	1.424	0.00	standard
Normal	364.6	<0.001	1.495	0.00	red
GARCH-t	17.9	0.007	1.175	0.18	green
FHS	9.9	0.129	1.157	1.00	green

- Only FHS passes the DQ test at 5% ($p = 0.129$); GARCH- t fails it ($p = 0.007$) through its excess of breaches
- FZ0 (lower is better): FHS and GARCH- t are not significantly different (DM $t = 1.39$), both beat HS and Normal ($|t| > 4.9$)
 - ▶ The 10% model confidence set is {GARCH- t , FHS}
- Comparative ES backtest: Normal red, GARCH- t and FHS green: the ranking of Step 4 survives formal inference

How Much Can a Backtest Detect? The Power of Kupiec



- Exact binomial probability that the 5% Kupiec test rejects a VaR 1% model, by its true breach rate
- $T = 250$ (one Basel year): a true rate of 2% is detected only 24% of the time; rejection needs 0 or at least 7 breaches
- Minimum detectable effect (MDE), here a true breach rate, with 80% power: 3.65% at $T = 250$, 2.05% at $T = 1000$, 1.45% at $T = 5432$
 - ▶ A true 1.3% is detected with probability 55% only
- A non-rejection with low power is “no evidence”, not “no problem”: state the power before the test

The Project in Four Steps

- ▣ An empirical study of one question about financial markets, answered with the methods of this course
 - ▶ **70%** of the final grade (quizzes and seminar activity 20%, attendance 10%, no final exam)
- ▣ **1. Replicate:** reproduce one full table or figure of a published paper (or of the lecture), cell by cell
 - ▶ Report the discrepancy of every cell and explain it; an AI assistant cannot guess a published table
- ▣ **2. Extend**
 - ▶ New data, a new market (BVB, crypto) or a more recent sample, under a **pre-analysis plan** recorded in a dated commit before the test period is opened
 - ▶ The plan states the null, the test, the loss, the sample split, the identification strategy, the power of the main test and the specification grid with its correction
- ▣ **3. Deliver on GitHub**
 - ▶ Code, a lockfile of package versions, fixed random seeds, data provenance; one notebook regenerates every number
 - ▶ [AI_USE.md](#) and [AI_ERRORS.md](#) are mandatory; the report is an 8–12-page paper
- ▣ **4. Present and defend:** team presentation, then a 10-minute oral defence
 - ▶ Each member explains one result and answers a “what changes if...” question; grades can differ between members

Replication Standard

- Target: one full table or figure, with its DOI, page and table number, not a single coefficient
 - ▶ Report a cell-by-cell comparison: published value, your value, difference, and the reason for each gap (sample, filter, data vendor, definition)
- Large-scale replications show why: many anomalies shrink out of sample or with better data [Hou, Xue & Zhang \(2020\)](#), while most factors replicate once the design is right [Jensen, Kelly & Pedersen \(2023\)](#)
- Reproducibility: a lockfile of package versions, fixed random seeds, series names and dates of the data, one command that regenerates every number
 - ▶ Different teams on the same data disagree more than their standard errors suggest [Menkveld et al. \(2024\)](#): more reproducible work is associated with a smaller gap (research case study, this chapter)

Choosing the Question

A good question is

- ▣ **Specific:** one market, one period, one quantity
- ▣ **Testable:** a null hypothesis and a benchmark
- ▣ **Feasible**
 - ▶ Data you can obtain and clean in a few weeks (the course data sets, in the course GitHub repository; public sources: BNR, FRED, Kenneth French Data Library, DefiLlama)
- ▣ **Relevant:** an answer matters to an investor, a bank or a regulator

Examples

- ▣ Too vague: "Is Bitcoin risky?"
 - ▶ Better: "Did spot ETFs change the one-day VaR 1% of Bitcoin?"
 - ▶ Then: "Which model forecasts it best since 2024?"
- ▣ Too broad: "Can machine learning predict the market?"
 - ▶ Better: "Does a boosting model beat log-HAR for next-day BET volatility, in QLIKE, 2018–2026?"

Methods: From Question to Chapter

If your question is about	Start from
Return distribution, tails, predictability	Chapters 1–2: moments, robust tests, variance ratios
Expected returns, factors, portfolios	Chapters 3–4: Fama–MacBeth, shrinkage, out-of-sample Sharpe tests
Volatility and correlation	Chapters 5, 6, 9: GARCH, DCC, copulas, log-HAR
Risk measures and capital	Chapters 7–8: FHS, EVT, backtests, FZ0
Liquidity, options, continuous time	Chapters 10–12: Amihud, Black–Scholes, VIX, variance risk premium
Forecasting with ML, deep learning or text	Chapters 13–15: purged CV, foundation models, sentiment
Crypto, bubbles, systemic risk	Chapters 16–18: stablecoins, GSADF, CoVaR, networks

- ▣ Starting points: the Part C question of every seminar (table in the [Appendix](#))
 - ▶ Your own question is welcome if it passes the five questions of this chapter

Identification: from Correlation to a Claim

- Event questions (“did spot ETFs, the FTSE upgrade or a bank tax change X?”) need a counterfactual
 - ▶ A control market not hit by the event: difference-in-differences on the risk measure
 - ▶ A synthetic control built from several untreated markets [Abadie, Diamond & Hainmueller \(2010\)](#); placebo event dates (e.g. $\pm$ one year)
- Unknown timing: a sup-Wald break test [Andrews \(1993\)](#); report the estimated break date, not an imposed one
- Forecasting questions: out of sample only, with a test period declared in the plan
- One event is $n = 1$: say what the design can and cannot identify

Power and Robustness without p-Hacking

- **Power:** before the main test, compute the smallest effect the design detects with 80% power at 5% (MDE)
 - ▶ Deliverable: a simulated power curve of the main test at your T
 - ▶ Short crypto, ETF or post-event samples make many questions underpowered (Kupiec at $T = 250$: MDE 3.65%)
- The pre-analysis plan is a timestamped commit in the repository, made before the test period is opened
- Specification curve: estimate every defensible combination of window, lags, α and sample start
 - ▶ Plot all estimates sorted, with their intervals [Simonsohn, Simmons & Nelson \(2020\)](#)
 - ▶ Report the spread across specifications, not the best cell
- Correct the family: Holm for any false discovery, Benjamini–Hochberg for their share, or White's bootstrap reality check for data snooping [White \(2000\)](#)
 - ▶ False and missed discoveries in finance [Harvey & Liu \(2020\)](#); the scientific outlook of the field [Harvey \(2017\)](#)

Forecast Evaluation for Risk Projects

- ▣ Volatility: compare forecasts with a noisy, conditionally unbiased proxy only through a robust loss (MSE or QLIKE) [Patton \(2011\)](#)
 - ▶ Many models: the model confidence set [Hansen, Lunde & Nason \(2011\)](#), not the lowest average loss
- ▣ VaR: coverage, independence and the DQ test [Engle & Manganelli \(2004\)](#); with a benchmark, conditional predictive ability tests [Giacomini & White \(2006\)](#)
 - ▶ State the power of every backtest at your T
- ▣ ES: score the pair (VaR, ES) with FZ0; comparative backtests with three zones [Nolde & Ziegel \(2017\)](#); dynamic (VaR, ES) models [Patton, Ziegel & Chen \(2019\)](#)
 - ▶ The integrated case study (Steps 1–5) is the template

The Report and the Presentation

The report: an 8–12-page paper

- **1. Introduction**
 - ▶ Question, hypothesis, contribution; **2. Literature:** 10–15 clickable references, at least 3 from the last five years, including Franke, Härdle & Hafner (2019), McNeil, Frey & Embrechts (2015)
- **3. Data and replication:** sources, period, cleaning rules; the replicated table cell by cell
- **4. Design:** plan, identification, power, benchmark, loss, tests
- **5. Results** with uncertainty; **6. Robustness:** the specification curve with the correction; **7. Conclusion:** the answer in one sentence, and its limits

The presentation (about 12 minutes)

- Question (1 slide)
 - ▶ Data and one chart that shows the problem (2)
 - ▶ Method and benchmark (2)
 - ▶ Main result with its uncertainty (2–3)
 - ▶ Robustness, limits, answer (2)
- One message per slide; readable axes, legend below the chart
- Every member presents and can answer questions on the code; then the 10-minute oral defence, without AI

Project Grading and Timeline

Component	What we look for
Replication (step 1)	A published table or figure reproduced cell by cell; every gap explained
Question, data, extension (step 2)	Specific, testable question; documented data and calendar alignment
Methods and inference	Plan, identification, power, real-time design, benchmark, specification curve with a correction
Interpretation and report	Economic meaning, limitations, honest conclusion; 8–12 pages, 10–15 references
GitHub repository (step 3)	One command reproduces every result (lockfile, seeds); AI_USE.md, AI_ERRORS.md
Presentation and defence (step 4)	Individual: one result explained, answers to “what changes if...”

- Week 1 after this chapter: team, question, one-page proposal (Seminar 19, C1), GitHub repository
 - ▶ Weeks 2–3: data, replication (step 1), plan recorded in a dated commit
- Weeks 4–5: extension, evaluation, robustness (step 2)
 - ▶ Final weeks: repository with README and the two AI files (step 3), presentation and oral defence (step 4)
 - ▶ Exact dates are announced at the course and seminar

The AI Policy and the Two Mandatory Files

Policy

- ▣ **Allowed:** AI assistants for code, debugging, literature search and writing
- ▣ **Declared:** undeclared use of AI counts as plagiarism
- ▣ **Checked:** you are responsible for every line; a reference that does not exist counts as fabricated data
- ▣ **No AI during the oral defence:** “the AI wrote it” is not an explanation

The two files

- ▣ **AI_USE.md:** tools and versions, who used them for what, where AI-written code or text was kept, how you verified the replicated table, data, formulas and references
- ▣ **AI_ERRORS.md:** at least **three** places where the assistant was wrong: what you asked, what it answered, what was wrong, how you found out, the correct version
- ▣ Errors you caught are a strength of the project, not a weakness

Typical AI Errors in Financial Modelling (1/2)

- ▣ **Look-ahead:** a rolling statistic or a scaler that includes day t in the forecast for day t
 - ▶ Check: change the last observation; the forecast for that day must not move; shift by one day (`.shift(1)`)
- ▣ **Sign of VaR:** the 1% quantile of returns reported as VaR, or the 99% quantile of returns (a gain) used as the loss
 - ▶ Check: $\text{VaR}_\alpha = -q_\alpha(r)$, with $q_\alpha(r)$ the α -quantile of returns, is positive whenever $q_\alpha(r) < 0$; a one-day VaR 1% of an index is a few percent
- ▣ **Annualisation:** volatility multiplied by 252 instead of $\sqrt{252}$, or 252 days used for crypto that trades every day
 - ▶ Check: over h periods, means scale with h , volatilities with $\sqrt{h}$ (i.i.d. returns); use the real number of trading days (365 for crypto)

Typical AI Errors in Financial Modelling (2/2)

▣ **Random cross-validation (CV) on time series**

- ▶ Shuffled K-fold trains on the future (69.4% accuracy on pure noise, [Chapter 13](#))
- ▶ Check: is any test date earlier than a training date? Use walk-forward or purged CV with an embargo [López de Prado \(2018\)](#)

▣ **Invented references**, wrong degrees of freedom, library defaults you did not choose, numbers stated without running code

- ▶ Check: search every DOI on [Crossref](#) and find the claim in the paper; every seminar has a C2 exercise that trains this

How Much Do Researchers' Choices Matter? (1/2)

- ▣ Menkveld et al. (2024), *Journal of Finance* 79(3): 164 research teams test the same six hypotheses on the same data (Finance Crowd Analysis Project, #fincap; Sec. I.A)
- ▣ **Standard error**
 - ▶ Uncertainty from sampling, the data-generating process
 - ▶ **nonstandard error**: uncertainty from the analysis path each researcher picks, the evidence-generating process (Introduction)
 - ▶ Model: team j reports $X_j = e_j + \varepsilon_j$, e_j the researcher fixed effect of its path, ε_j the sampling error (Sec. I.C, eq. 1)
 - ▶ Nonstandard error $:= Q_{0.75}(x) - Q_{0.25}(x)$, the interquartile range (IQR) of the estimates across teams (eq. 3), $Q_p(x)$ being the p -quantile of the team estimates x
 - ▶ The IQR, not the standard deviation, because the e_j can have fat tails

How Much Do Researchers' Choices Matter? (2/2)

▣ Data

- ▶ All trades in the Euro Stoxx 50 index futures, 2002–2018, 720 million records
 - ▶ Six hypotheses of “no annual trend” (market efficiency, realised spread, client share of volume, ...)
 - ▶ Each team reports the average yearly change in % with its standard error (SE)
 - ▶ Four stages in 2021: independent analysis, peer feedback, the five best papers shared, final estimates; design details in the [▶ Appendix](#)
- ▣ Link to this chapter: every team project is one path through the “garden of forking paths”
- ▶ Pre-registration, the specification grid and a repository that reruns every number are the remedies

Key Results: Size and Quality

▣ Size

- ▶ For market efficiency the median estimate is -1.1% per year, with an IQR of 6.7 percentage points (pp)
- ▶ Even for the client share of volume, a simple ratio, the IQR is 1.2 pp around -3.3% (Table I, Panel C)
- ▶ The median SE of 2.5% implies a Gaussian IQR of $1.35 \times 2.5 = 3.4\%$ (the IQR of a Normal distribution is 1.35 standard deviations), half the 6.7 pp
- ▶ The no-dispersion null is rejected for all six hypotheses, $p < 0.0001$ (Table II)

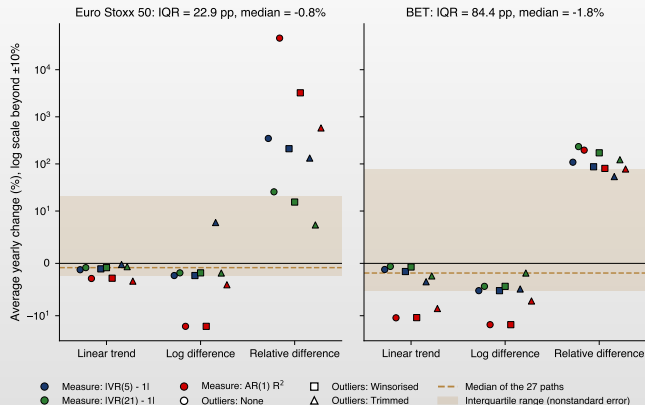
▣ Quality

- ▶ A rise of one standard deviation in reproducibility is associated with a 25.0% lower IQR, the same rise in peer rating with a 33.3% lower IQR (Table III, Fig. 2)
- ▶ Code that reruns end to end is associated with answers closer to the consensus

Key Results: Feedback, Beliefs and Forks

- ▣ **Feedback:** from stage 1 to stage 4 the IQR falls by 47.2% and the interdecile range by 68.2% (Table IV, Fig. 3)
 - ▶ Peer review narrows the dispersion but does not remove it
- ▣ **Beliefs:** pooled over hypotheses, the teams' predicted standard deviation of estimates is 71.7% below the realised one (Sec. II.B.3, Fig. 4)
 - ▶ Researchers underestimate how much teams analysing the same data can differ
- ▣ **Forks**
 - ▶ Model choice and sampling frequency move the estimates most
 - ▶ A relative-change model gives a median annual trend of $-23\,000\%$ on daily data, -200% monthly and -4.56% yearly (Fig. 6, Fig. 7)
 - ▶ Jensen's inequality for relatives $M_t = X_t/X_{t-1}$: $\mathbb{E} \ln M_t \leq \ln \mathbb{E} M_t$
 - ▶ Log differences add up exactly across frequencies and a linear trend is linear in the data, so neither has this frequency wedge

A Mini-Multiverse on Course Data: 27 Paths per Index (1/2)

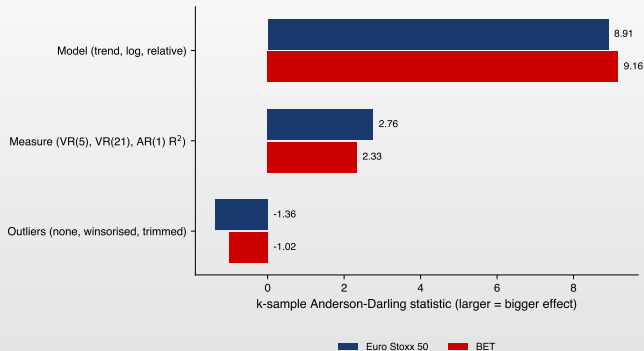


- Daily closes 2002–2018 from the course data
 - ▶ Forks of Table V: outliers $\times$ measure $\times$ model, 27 paths
 - ▶ The band is the IQR, the nonstandard error of the multiverse
- IQR: 22.9 pp (Euro Stoxx 50) and 84.4 pp (BET)
 - ▶ Around medians of -0.8% and -1.8% a year
- Measures of inefficiency
 - ▶ $|VR(q) - 1|$ ($VR(q)$: the variance ratio of Chapter 2) and R^2 : departures from a random walk
 - ▶ A positive trend means declining efficiency

A Mini-Multiverse on Course Data: 27 Paths per Index (2/2)

- The model fork creates the dispersion
 - ▶ Over the nine linear-trend paths the IQR is only 2.0 pp and 7.5 pp
 - ▶ Relative differences explode: medians +210.4% and +108.7%, up to 46 823%
 - ▶ The paper reports the same explosion (Fig. 7)
- Sampling benchmark: $1.35\times$ the median SE (25.9 and 50.8 pp)
 - ▶ 1.35 standard deviations: the IQR of a Normal distribution
 - ▶ The IQR exceeds this benchmark for the BET, not for the Euro Stoxx 50

Which Fork Moves the Estimates? (1/2)



- *k*-sample Anderson–Darling statistic of the 27 estimates, grouped by the option at each fork (Sec. II.C)
 - ▶ It compares the distributions of the *k* groups; larger means more separated
- Euro Stoxx 50 and BET
 - ▶ Model 8.91 and 9.16; measure 2.76 and 2.33
 - ▶ Outliers −1.36 and −1.02
- Same ranking as the paper (Fig. 6): model choice moves the estimates most

Which Fork Moves the Estimates? (2/2)

- A descriptive ranking, not a test
 - ▶ The 27 estimates share the same data
 - ▶ The standardised statistic is centred at zero when the distributions are equal
 - ▶ A negative value means less separation than chance alone, not evidence of no effect
- Holds on a large euro-area index and on a frontier-market index
- For your project
 - ▶ Fix the model of the yearly change in the plan (log differences or linear trends)
 - ▶ Report the whole grid and the fork that drives it

Limits and an Open Extension

▣ Limits stated by the authors

- ▶ The multiverse covers a few forks only and explains the reported estimates poorly (footnote 32)
- ▶ Beliefs were asked as standard deviations, not IQRs (footnote 28)
- ▶ The test of dispersion assumes consistent reported SEs (Sec. I.C); the dynamics of journal refereeing are kept out of scope (Sec. I.A)
- ▶ A multiverse spans only the options in its grid; its IQR neither estimates nor bounds from below the dispersion across real teams

▣ Data limitation of the replication

- ▶ Daily closes only, so the intraday frequency pairs of Table V cannot be run (Seminar 19, C3 extends it to the last full year)

▣ Open extension: does a pre-registered grid shrink the nonstandard error of a risk question?

- ▶ Measure the IQR of VaR 1% conclusions for the BET across defensible paths
- ▶ For your project: report the distribution over your grid and the forks that move it most, not a single cell

An Open Question

- ▣ **Open question:** how much of a result is the data, and how much the researcher?
 - ▶ Replication is not hopeless: in a Bayesian model of replication most asset-pricing factors replicate and work out of sample in 93 countries [Jensen, Kelly & Pedersen \(2023\)](#)
 - ▶ AI assistants make every extra specification almost free: the forking paths multiply, but so does the ability to report all of them
 - ▶ Formal version: the conclusion must hold over a pre-registered set $\mathcal{S}$ of defensible specifications, with the family corrected (FWER or FDR)
 - ▶ Report its distribution over $\mathcal{S}$, not the best cell

The Discovery Loop and Its Checks

□ Loop

- ▶ Question and H_0/H_1 (week 1)
- ▶ Literature and the replication target seen in the PDF (weeks 1–2)
- ▶ Pre-registration recorded in a dated commit before the first result (week 2)
- ▶ Replication as a passing test in the repository (weeks 2–3)
- ▶ The whole grid (weeks 4–5)
- ▶ An AI examiner before the defence
- ▶ “Here are all the cells we ran. Which conclusions survive a Holm correction, and which would a referee call data snooping?”

□ Checks by the researcher

- ▶ The target number is in the paper
- ▶ The plan precedes the first results file in the commit history
- ▶ Every run is logged and counted
- ▶ A fresh clone regenerates every number
- ▶ Each member can derive the key formula without AI
- ▶ Prompts for every step and the test for each check:

▶ Appendix

Mini-Case: Replicate, Then Extend

- **Replicate:** FHS VaR 1% for the BET; [Chapter 8](#) reports 59 breaches in 4 937 days (1.20%, Kupiec $p = 0.181$)
 - ▶ The case study of this chapter: 67 in 5 432 days (1.23%, $p = 0.095$)
 - ▶ Same model, different evaluation start (2007 vs 2005) and settings: document every gap before calling a replication failed
- A typical assistant proposal, reading each p -value alone at 5%: “only GARCH-EVT is correctly calibrated on all four assets”
- **Extend:** the 24 conditional-coverage tests of [Chapter 8](#) (4 assets $\times$ 6 models) form one family
 - ▶ Rejections at 5%: 12 without correction, 10 with Holm (FWER), 11 with BH (FDR 5%)
 - ▶ FHS fails only on the S&P 500 ($p = 0.040$)
 - ▶ Within the family of 24 it survives, so after Holm FHS and GARCH-EVT pass on all four assets
- Interpretation: “FHS fails for the S&P 500” depends on whether the test is read alone or in its family; pre-register the family and report both

Your Turn: the Capstone Plan

- ▣ **Question:** your project idea from Part C, written as H_0/H_1 , with the observation that would falsify it
- ▣ **Replicate first:** one published table seen in the paper, turned into a passing test in the repository
 - ▶ Then extend over a pre-registered grid, with the family corrected and the full distribution reported
- ▣ **Deliverables** along the timeline: proposal and replication target, pre-registration commit, passing replication test, grid of results, mock defence, report, oral defence
- ▣ **AI log:** the prompts that changed the design in `AI_USE.md`; every caught error, with the test that caught it, in `AI_ERRORS.md` ([templates](#))

Key Takeaways

- ▣ Returns are hard to predict; their risk is not: volatility, tails and dependence can be forecast
- ▣ Every model is judged out of sample, against a simple benchmark, with a loss that fits the target
- ▣ Heavy tails change inference: use robust errors and the bootstrap, report intervals, state the power
- ▣ Most wrong results come from the design, not from the model: look-ahead, overfitting, multiple testing, calendars
- ▣ The analysis path is itself a source of uncertainty: pre-register, replicate, report the whole grid
- ▣ New tools (ML, foundation models, LLMs) raise the bar for evaluation, not lower it

Check Yourself and Further Study

Retrieval questions

- Why is the sample kurtosis of BET returns not an estimate of anything?
- Why can the Wald interval for GARCH persistence not be reported?
- What replaces it?
- What does a non-rejection of the Kupiec test at $T = 250$ tell you?
- Why is the IQR of a 27-path multiverse not the nonstandard error of 164 teams?

Further study

- Textbooks: Franke, Härdle & Hafner (2019); Härdle, Chen & Overbeck (2017); McNeil, Frey & Embrechts (2015); Tsay (2010)
- Advanced courses on quantinar.com:  Statistics of Financial Markets
 - ▶  Measuring Statistical Risk
- Code and data: [Quantlets/Ch_19 \(GitHub\)](#)

Appendix: Key Inference by Chapter (1/17)

□ Chapter 0, Financial Markets in 2026

- ▶ Data quality is a modelling decision: source, adjustment and calendar change the result
- Mean and volatility estimated from i.i.d. log returns, annualised with the number of periods a year

$$\text{se}(\hat{\mu}_{\text{ann}}) = \frac{\sigma_{\text{ann}}}{\sqrt{Y}}, \quad \text{se}(\hat{\sigma}) \approx \sigma \sqrt{\frac{K-1}{4T}}$$

- ▶ Se: standard error
- ▶ Y: sample length in years, whatever the frequency
- ▶ T: number of daily returns
- ▶ σ , σ_{ann} : daily and annualised volatility
- ▶ K: population kurtosis (a finite fourth moment is needed)
- ▶ Serial dependence calls for long-run variances
- SPY 2015–2026: $\text{se}(\hat{\mu}_{\text{ann}}) \approx 5.1\%$ for a mean of 13.7%: means are barely estimable, volatilities are

Appendix: Key Inference by Chapter (2/17)

Chapter 1, Stylised Facts and Returns

- ▶ Heavy tails and volatility clustering in every market [Cont \(2001\)](#)

Ljung–Box test of no autocorrelation up to lag m

$$Q(m) = T(T+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k} \sim \chi_m^2$$

- ▶ $\hat{\rho}_k$: sample autocorrelation at lag k
- ▶ T : sample size
- ▶ The χ_m^2 reference needs i.i.d. returns
- ▶ Under volatility clustering it over-rejects
- ▶ Use the robust Q^* with summands $T\hat{\rho}_k^2/\hat{\tau}_k$, $\hat{\tau}_k$ the estimated variance factor of $\sqrt{T}\hat{\rho}_k$ (about 1 for i.i.d. returns) [Lobato, Nankervis & Savin \(2001\)](#)

Tails: Hill estimator $\hat{\alpha}$ of the tail index, se $\approx \hat{\alpha}/\sqrt{k}$ [Hill \(1975\)](#)

- ▶ α : the probability of a loss above x decays like $x^{-\alpha}$; k : number of largest losses used
- ▶ If $\alpha < 4$ the sample kurtosis is not consistent and its bootstrap fails [Athreya \(1987\)](#)

Appendix: Key Inference by Chapter (3/17)

□ Chapter 2, Market Efficiency

- ▶ Efficiency varies over time (adaptive markets); tests need robust inference

□ Heteroskedasticity-robust variance ratio test [Lo & MacKinlay \(1988\)](#)

$$VR(q) = \frac{\text{Var}(r_t + \dots + r_{t-q+1})}{q \text{Var}(r_t)}, \quad z^*(q) = \frac{\widehat{VR}(q) - 1}{\widehat{se}^*} \Rightarrow N(0, 1)$$

- ▶ q : horizon in periods; $VR = 1$ under a random walk, $VR > 1$: positive autocorrelation, $VR < 1$: mean reversion
- ▶ $\widehat{se}^*$: heteroskedasticity-robust standard error; $z^* \Rightarrow N(0, 1)$ under a martingale difference null

- Several horizons q at once: the joint test $\max_q |z^*(q)|$ with studentised maximum modulus critical values [Chow & Denning \(1993\)](#)
 - ▶ Rolling windows need a multiplicity correction too

Appendix: Key Inference by Chapter (4/17)

Chapter 3, Factor Models and Asset Pricing

- ▶ Hundreds of published factors require higher hurdles [Harvey, Liu & Zhu \(2016\)](#)
- GRS test that all pricing errors (intercepts) are zero [Gibbons, Ross & Shanken \(1989\)](#)

$$\text{GRS} = \frac{T - N - K}{N} \frac{\hat{\alpha}^\top \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \hat{\mu}_f^\top \hat{\Omega}^{-1} \hat{\mu}_f} \sim F(N, T - N - K)$$

- ▶ T periods, N assets, K factors; $\hat{\alpha}$: the N intercepts (alphas); $\hat{\mu}_f$: factor means; $\hat{\Sigma}$, $\hat{\Omega}$: residual and factor covariances, divisor T
- ▶ Exact F only under i.i.d. Normal errors; a large GRS means the factors do not price the assets
- Fama–MacBeth s.e. ignore that betas are estimated: Shanken's covariance for estimated factor loadings [Shanken \(1992\)](#)

$$\hat{V}_{\text{sh}} = (1 + c)(\hat{V}_{FM} - \hat{\Sigma}_f/T) + \hat{\Sigma}_f/T, \quad c = \hat{\gamma}^\top \hat{\Sigma}_f^{-1} \hat{\gamma}$$

- ▶ $\hat{\gamma}$: estimated factor risk premia
- ▶ $\hat{\Sigma}_f$: factor covariance
- ▶ $\hat{V}_{FM}$: Fama–MacBeth covariance of $\hat{\gamma}$
- ▶ $c \geq 0$ inflates it, more when premia are large relative to factor risk

Appendix: Key Inference by Chapter (5/17)

□ Chapter 4, Portfolio Optimisation

- ▶ Shrinkage, constraints and risk-based rules stabilise the weights; 1/N is a hard benchmark [DeMiguel, Garlappi & Uppal \(2009\)](#)
- Standard error of the Sharpe ratio $SR = (\mu - r_f)/\sigma$ per period, i.i.d. Normal returns [Lo \(2002\)](#)

$$se(\widehat{SR}) \approx \sqrt{(1 + \widehat{SR}^2/2)/T}$$

- ▶ μ , σ : mean and volatility per period; r_f : risk-free rate; T : number of periods; serial correlation changes the annualisation factor
- Is portfolio A better than 1/N? Test $SR_A - SR_B = 0$ with HAC or studentised bootstrap inference [Ledoit & Wolf \(2008\)](#), not by comparing two point estimates

Appendix: Key Inference by Chapter (6/17)

□ Chapter 5, Conditional Volatility: GARCH

- ▶ Persistence close to 1; asymmetry in equities; heavy-tailed innovations

□ QMLE (quasi-maximum likelihood) of GARCH: the sandwich covariance [Bollerslev & Wooldridge \(1992\)](#)

$$\sqrt{T}(\hat{\theta} - \theta) \Rightarrow N(0, A^{-1}BA^{-1})$$

- ▶ $\theta = (\omega, \alpha, \beta)$: parameters of $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$; A : expected Hessian of the Normal log-likelihood (sign changed); B : variance of its score
- ▶ $A = B$ when the innovations are Normal; Hessian-only s.e. (A^{-1}) are wrong for non-Normal innovations

□ Near $\alpha + \beta = 1$ the parameter is close to the boundary of the covariance-stationary region: Wald intervals are unreliable

- ▶ Use profile likelihood or bootstrap restricted to the parameter space [Andrews \(1999\)](#)

Appendix: Key Inference by Chapter (7/17)

□ Chapter 6, Multivariate Volatility and Dependence

- ▶ Copulas separate the margins from dependence; tail dependence drives joint losses
- DCC is estimated in two steps: second-step s.e. must account for the first-step volatilities [Engle \(2002\)](#)
- Crisis vs calm correlation: ρ rises mechanically with volatility; test on the adjusted correlation [Forbes & Rigobon \(2002\)](#)

$$\rho^* = \frac{\rho}{\sqrt{1 + \delta(1 - \rho^2)}}$$

- ▶ ρ : correlation measured in the crisis; δ : relative rise in the variance of the source market from calm to crisis; $\rho^* < \rho$ when $\delta > 0$

Appendix: Key Inference by Chapter (8/17)

□ Chapter 7, Value-at-Risk and Expected Shortfall

► Filtered historical simulation and conditional EVT react to volatility in days McNeil & Frey (2000)

□ Historical VaR, i.i.d. returns with a continuous density $f(q_\alpha) > 0$

$$\sqrt{T}(\hat{q}_\alpha - q_\alpha) \Rightarrow N(0, \alpha(1 - \alpha)/f(q_\alpha)^2)$$

► q_α : the α -quantile of returns, $\text{VaR}_\alpha = -q_\alpha$; $\hat{q}_\alpha$: the empirical quantile; f : return density; in the tail f is small, so the estimate is noisy

► Under dependence $\alpha(1 - \alpha)$ becomes the long-run variance of the hits $1\{r_t \leq q_\alpha\}$ (1 if the condition holds, 0 otherwise)

□ ES averages the αT largest losses: with αT of a few dozen and a tail index near 3, its variance is dominated by one or two days

Appendix: Key Inference by Chapter (9/17)

□ Chapter 8, Backtesting and Evaluating Risk Forecasts

- ▶ Rank (VaR, ES) forecasts with a consistent score (FZ0) and Diebold–Mariano tests [Fissler & Ziegel \(2016\)](#)
- VaR backtests: the number and the timing of breaches (formulas: Step 4 of the case study)
 - ▶ Kupiec $LR_{uc} \sim \chi^2_1$ (is the breach rate α ?); Christoffersen $LR_{ind} \sim \chi^2_1$ (is a breach independent of yesterday's?)
 - ▶ DQ test: regress $I_t - \alpha$ on lagged hits and VaR_t, $I_t = 1$ if the loss exceeds VaR_t; $DQ \sim \chi^2_k$, k the number of regressors [Engle & Manganelli \(2004\)](#)
- Power is low: at $T = 250$ the 5% Kupiec test detects a true rate of 2% only 24% of the time
 - ▶ Compare models with conditional predictive ability tests [Giacomini & White \(2006\)](#)

Appendix: Key Inference by Chapter (10/17)

□ Chapter 9, Realised Volatility

- ▶ Returns divided by realised volatility have a distribution close to the Normal distribution

□ HAR errors are heteroskedastic and, for multi-day targets that overlap, serially correlated: use HAC s.e.

- ▶ Forecasts are compared with DM tests on a robust loss [Patton \(2011\)](#)

□ Microstructure noise biases RV (the sum of squared intraday returns) upwards at high frequency: two-scale estimators remove the bias [Zhang, Mykland & Aït-Sahalia \(2005\)](#)

□ Chapter 10, Market Microstructure

- ▶ Liquidity follows the clock and the VIX, and disappears in stress

□ The Roll spread: the bid–ask bounce makes consecutive price changes negatively correlated

$$s_{\text{Roll}} = 2\sqrt{-\text{Cov}(\Delta p_t, \Delta p_{t-1})}$$

- ▶ Δp_t : change of the transaction price; when the sample covariance is positive the estimator is undefined, so report how often

□ An illiquidity premium is a cross-sectional claim: Fama–MacBeth with Newey–West s.e. and the Shanken correction [Shanken \(1992\)](#)

Appendix: Key Inference by Chapter (11/17)

□ Chapter 11, Continuous-Time Models

- ▶ Jumps explain the tails, stochastic volatility the clustering
- Normality of GBM log returns: Jarque–Bera $JB = \frac{T}{6} (S^2 + (K - 3)^2/4) \sim \chi^2_2$ rejects everywhere
 - ▶ S : sample skewness; K : sample kurtosis; both 0 and 3 under the Normal distribution
 - ▶ Jumps: $(RV - BV)$ standardised $\Rightarrow N(0, 1)$ [Barndorff-Nielsen & Shephard \(2006\)](#)
 - ▶ BV : bipower variation, a sum of products of adjacent absolute intraday returns, robust to jumps
- The test is daily: over many days, control the number of false jump days

Appendix: Key Inference by Chapter (12/17)

- **Chapter 12, Options and the Volatility Surface**

- ▶ Smile and skew encode the fear of crashes

- The variance risk premium

$$VRP_t = VIX_t^2 - RV_{t,t+21}$$

- ▶ VIX_t^2 : the risk-neutral expected variance over the next month; $RV_{t,t+21}$: realised variance over the next 21 trading days
- ▶ Consecutive values overlap, so the mean needs Newey–West s.e. with at least 21 lags

- Implied volatility is a risk-neutral forecast: the premium, not a forecast error, explains most of the gap

Appendix: Key Inference by Chapter (13/17)

□ Chapter 13, Machine Learning in Finance

- ▶ Purged cross-validation, meta-labelling and deflated Sharpe ratios [Bailey & López de Prado \(2014\)](#)
- Selection among N strategies: the deflated Sharpe ratio tests $SR > E[\max_N \widehat{SR}]$ under the null of no skill [Bailey & López de Prado \(2014\)](#)
 - ▶ $E[\max_N \widehat{SR}]$: the expected best estimated Sharpe ratio among N strategies whose true Sharpe ratio is zero; it grows with N
- Cross-validation on time series must purge overlapping labels and embargo the test fold, or its accuracy is biased upwards

Appendix: Key Inference by Chapter (14/17)

- **Chapter 14, Deep Learning and Time-Series Foundation Models**

- ▶ Evaluate them with [Chapter 8](#) tools, after the release of their weights

- DM test on QLIKE differences with HAC variance [Diebold & Mariano \(1995\)](#)

$$t_{DM} = \frac{\bar{d}}{\sqrt{\hat{\Omega}/T}} \Rightarrow N(0, 1)$$

- ▶ d_t : QLIKE of model 1 minus QLIKE of model 2 on day t , $\bar{d}$ its mean over T days
- ▶ $\hat{\Omega}$: HAC long-run variance of d_t
- ▶ Valid for non-nested comparisons

- With many candidate models, the model confidence set (MCS) controls the family-wise error [Hansen, Lunde & Nason \(2011\)](#)

Appendix: Key Inference by Chapter (15/17)

□ Chapter 15, LLMs and Sentiment Analysis

- ▶ Text embeddings enter forecasting models as features

□ Predictive regression $r_{t+1} = a + bS_t + \varepsilon_{t+1}$

- ▶ S_t : sentiment score of day t
- ▶ b : predictive slope
- ▶ A persistent S_t whose innovations correlate with returns biases $\hat{b}$ and the t -test [Stambaugh \(1999\)](#)

□ Classifier accuracies on the same texts are paired: compare them with the exact McNemar test, not two separate intervals

□ Chapter 16, Digital Assets and DeFi

- ▶ Stablecoins, automated market makers and spot ETFs connect crypto to mainstream finance

□ Crypto factor models: GRS and Fama–MacBeth with the Shanken correction [Gibbons, Ross & Shanken \(1989\)](#), [Shanken \(1992\)](#); heavy tails call for HAC or bootstrap versions

□ Tails: Hill indices with intervals, not sample kurtosis ([Chapter 1](#))

- ▶ A break in a correlation is tested with an unknown date: the sup-Wald statistic is the largest Wald statistic over candidate break dates [Andrews \(1993\)](#)

Appendix: Key Inference by Chapter (16/17)

□ Chapter 17, Bubbles and Crashes

- ▶ Right-tailed unit-root tests date the start and the end of explosive episodes

□ Bubble tests: the largest right-tailed ADF statistic over many windows

$$GSADF = \sup_{r_2 \in [r_0, 1]} \sup_{r_1 \in [0, r_2 - r_0]} ADF_{r_1}^{\prime 2}$$

- ▶ $ADF_{r_1}^{\prime 2}$: the ADF (augmented Dickey–Fuller) statistic on the sample fraction $[r_1, r_2]$; r_0 : minimum window; SADF fixes $r_1 = 0$
- ▶ Under the unit-root null, SADF and GSADF converge to functionals of Brownian motion; critical values depend only on r_0 Phillips, Shi & Yu (2015)

□ Volatility shifts distort the size: use wild-bootstrap critical values; control the multiplicity of date-stamping

Appendix: Key Inference by Chapter (17/17)

□ Chapter 18, Systemic Risk and Networks

- ▶ CoVaR (the VaR of the system conditional on the distress of one institution), SRISK and spillover networks measure the contribution of each institution

- ΔCoVaR: the change in the system's VaR when institution i moves from its median to distress

$$\Delta\text{CoVaR}^i = \text{CoVaR}_{\alpha}^{\text{sys}|X^i=-\text{VaR}_{\alpha}^i} - \text{CoVaR}_{\alpha}^{\text{sys}|X^i=\text{median}}$$

- ▶ X^i : return of institution i
- ▶ $\text{CoVaR}_{\alpha}^{\text{sys}|i}$: VaR α of the system return given the state of i
- ▶ Differences between banks: paired tests on joint block-bootstrap draws with a Holm correction
- ▶ At $\alpha = 1\%$ only about 40 tail days, so extremal-quantile issues

- SRISK: LRMES (the expected loss of a bank when the market falls by more than 10% over 22 days) simulated from GJR-GARCH + DCC [Brownlees & Engle \(2017\)](#)

- ▶ Connectedness tables need bootstrap intervals and shrinkage vector autoregressions (VARs)

Appendix: Project Ideas: Part C of Every Seminar (1/2)

Ch.	Open question (link to the seminar)
0	Did spot ETFs change Bitcoin's correlation with equities?
1	Did BET-TR autocorrelation fall after the FTSE reclassification?
2	Did the BVB become more efficient after the 2020 FTSE upgrade?
3	Do factor premia exist on the Bucharest Stock Exchange?
4	An optimal portfolio of BVB blue chips in RON
5	Is the Romanian market's volatility driven by the US?
6	Did Romanian and euro-area equities become more integrated after 2020?
7	Capital for a BVB portfolio vs an S&P 500 portfolio

- Each seminar gives the data, a starting method and a reference analysis discussed in class

Appendix: Project Ideas: Part C of Every Seminar (2/2)

Ch.	Open question (link to the seminar)
8	Which risk model would you certify for a Romanian bank's trading book?
9	Is Bitcoin's volatility more predictable than the S&P 500's?
10	Does illiquidity carry a premium on the BVB?
11	Which continuous-time model fits BET and Bitcoin?
12	Is the variance risk premium a tradable signal?
13	Does a foundation model beat boosting?
14	Can a zero-shot foundation model replace GARCH for the BET?
15	Does headline sentiment pay after costs?
16	Is Bitcoin an alternative asset, a risk asset or a hedge?
17	Is there an AI bubble?
18	How exposed are Romanian banks to euro-area banking stress?

□ Chapter 20 (realised volatility with signatures, VOLARE data) also suits a forecasting project

Appendix: Turning an Idea into a Project

- Replicate first: one published table or figure from a paper behind the idea (step 1)
- Narrow it: one market, one period, one number to forecast or test
 - ▶ Example: Seminar 8 becomes “Which of HS, FHS and GARCH-EVT gives a correct VaR 1% for a 50/50 portfolio of the BET and EUR (EUR/RON exchange rate) in 2015–2026?”
- Extend it: a second market, a new model, a longer or more recent sample
 - ▶ Example: add a foundation model ([Chapter 14](#)) as a volatility filter inside FHS
- Keep the reference analysis as your benchmark and try to improve on it honestly

Appendix: Data Conventions for the Project

- Use the course data where possible: daily series of indices, ETFs, BVB stocks, crypto and FX, plus intraday bars
 - ▶ Public sources: BNR reference rates, FRED, Kenneth French Data Library, DefiLlama; the BVB archive for Romanian indices
- Apply the course conventions
 - ▶ Adjusted prices for ETFs and stocks, closing values for indices, FX and crypto
 - ▶ Remove holiday rows where the vendor repeated the previous price, and weekend quotes where the market is closed; annualise with the real frequency
 - ▶ Several assets: align prices on common trading days first, then compute returns
- Look for erroneous prices (bad ticks) before modelling: jumps of $\pm 50\%$ in adjusted prices are usually data errors

Appendix: Honest Evaluation Checklist

Before you look at the results

- ☐ Write the hypothesis, the benchmark and the loss function
- ☐ Fix the estimation window and the test period
- ☐ Decide how many models and parameters you will try, and the correction for them
- ☐ Write the identification strategy and the power of the main test

When you report

- ☐ Out-of-sample results only for claims about forecasting
- ☐ Every headline number with a standard error, CI or p -value
- ☐ Costs for trading strategies; subperiods and a second market for robustness
- ☐ What did **not** work, and why



Appendix: Where Projects Usually Lose Points

- ▣ A question that changes after the results are known
- ▣ In-sample fit presented as forecasting power
- ▣ No benchmark, or a weak one chosen to be beaten
- ▣ Numbers without uncertainty; many tests without correction
- ▣ Charts without units, legends inside the data, unexplained acronyms
- ▣ Code that does not run, or numbers in the report that the code does not produce
- ▣ AI output kept without checks: undeclared use, unchecked code, references that do not exist

Appendix: Nonstandard Errors, Data and Method (1/2)

□ Data

- ▶ All trades in the Euro Stoxx 50 index futures, 2002–2018, 720 million records, each side flagged as client or own account (Sec. I.A)
- ▶ Six hypotheses, each “no annual trend” in: market efficiency, realised bid–ask spread, client share of volume, realised spread on client orders, share of market orders among client orders, gross trading revenue of clients
- ▶ Each team chooses its own measure and reports the average yearly change in % with its standard error (SE)

□ Four stages in 2021: independent analysis (paper and code)

- ▶ Feedback from two peer evaluators
- ▶ The five best papers shared
- ▶ Final estimates without the code constraint (Sec. I.A)
- ▶ A pre-analysis plan was filed publicly before the experiment started (Sec. I.A)

Appendix: Nonstandard Errors, Data and Method (2/2)

▣ Quality

- ▶ Team quality (first principal component of five proxies), reproducibility score 0–100 by the certification agency Cascad, demeaned peer rating (Sec. I.B)
- ▶ Quantile regressions of the estimates at the 10th, 25th, 50th, 75th and 90th percentiles; a shrinking IQR means less dispersion (Sec. I.C)

▣ Inference

- ▶ Each estimate is tested against the median estimate, with a Bonferroni correction at a family level of 0.5% (Sec. I.C)
- ▶ Multiverse analysis: the teams report their choice at each fork of Table V
- ▶ The k -sample Anderson–Darling statistic ranks the forks by how much they move the estimates (Sec. II.C)

Appendix: Nonstandard Errors, Replication Details

▣ Forks from Table V

- ▶ Outliers (none; the annual measure winsorised or trimmed at its 2.5 and 97.5 percentiles), measure (variance ratio day to week, variance ratio day to month, R^2 of an AR model), model (linear trend, log difference, relative difference)
- ▶ Daily data allow the day-to-week and day-to-month comparisons ($VR(5)$, $VR(21)$); the intraday measure frequencies of Table V need intraday data

▣ Why relative differences explode

- ▶ $m_t = |VR - 1|$ or R^2 is close to zero in some years; the paper finds the same explosion, a median of $-23\,000\%$ on daily data (Fig. 7)
- ▶ No trend, stationary m_t : the linear slope and the mean log change are centred at zero, but for independent m_t , $\mathbb{E}(m_t/m_{t-1}) = \mathbb{E}m \cdot \mathbb{E}(1/m) \geq 1$ (Jensen), so the mean relative change is positive

▣ Anderson–Darling statistic

- ▶ The standardised statistic is centred at zero for equal distributions of independent samples
- ▶ The 27 estimates are strongly correlated, so tabulated critical values and p -values do not apply
- ▶ Paper, 164 teams: 6.7 pp against 3.4 pp (Table I, Panel C); in the mini-multiverse the IQR exceeds this benchmark for the BET, not for the Euro Stoxx 50

Appendix: The Discovery Loop, Prompts (1/2)

- **Week 1, the question:** sharpen a project idea into H_0/H_1 with a replication target
 - ▶ “Here is our project idea from Part C of Seminar N. Give three sharper, testable versions, the published table each would replicate and the series from the course data it needs.”
- **Weeks 1–2, literature and target:** the number to replicate is one you have seen in the PDF
 - ▶ “List the most cited and the most recent papers on this question with DOI; for each, the headline estimate, its table and its sample.”
 - ▶ Resolve every DOI on [Crossref](#); search with Semantic Scholar, Elicit or Google Scholar; open the table yourselves
- **Week 2, hypotheses and pre-registration,** recorded in a dated commit before the first result
 - ▶ “For our H_0 and H_1 , list competing hypotheses and the implication that separates each; then draft a pre-registration: sample, filters, windows, tests, correction.”
 - ▶ The commit history of the repository is the evidence that the plan came first

Appendix: The Discovery Loop, Prompts (2/2)

- **Weeks 2–3, code and replication:** the replicated table becomes a test in the repository
 - ▶ “Write a function that reproduces every cell of the published table from the course data and an assertion with the tolerance we accept.”
- **Weeks 4–5, extension and robustness:** run the whole pre-registered grid
 - ▶ “Here are all the cells we ran. Which conclusions survive a Holm correction, and which ones would a referee call data snooping?”
- **Before the defence, critique:** the assistant asks, the team answers
 - ▶ “Act as the examiner of our oral defence: ask the ten hardest questions on identification, inference and data. Do not answer them.”
 - ▶ The team answers without AI, as at the defence itself
- **Report:** the distribution of results over the grid, not the best cell; prompts in AI_USE.md, caught errors in AI_ERRORS.md

Appendix: What the Researcher Must Check

- **The replication target exists:** number, table and sample taken from the paper, not from the assistant's summary
 - ▶ Test: the report cites the page and the table; a second member finds the same number there
- **The plan came first**
 - ▶ Test: in the commit history, the pre-registration precedes the first results file
- **Forking paths:** more specifications tried than reported
 - ▶ Test: every run writes its configuration and result to a log; the report states their number and corrects the family
- **Reproducibility:** numbers entered manually in the report
 - ▶ Test: a fresh clone and one command regenerate every number to the reported digit
- **Understanding:** each member can derive the key formula and defend every choice
 - ▶ Test: an internal mock defence, without AI, one week before the real one

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